

Matrix Operations

Now that we've learned how to create a matrix, let's learn how to use functions and perform operations on it!

Run the following code to create the `stock.matrix` from earlier

In [18]:

```
# Prices
goog <- c(450,451,452,445,468)
msft <- c(230,231,232,236,228)

# Put vectors into matrix
stocks <- c(goog,msft)
stock.matrix <- matrix(stocks,byrow=TRUE,nrow=2)

# Name matrix
days <- c('Mon','Tue','Wed','Thu','Fri')
st.names <- c('GOOG','MSFT')
colnames(stock.matrix) <- days
rownames(stock.matrix) <- st.names

# Display
stock.matrix
```

Out[18]:

	Mon	Tue	Wed	Thu	Fri
GOOG	450	451	452	445	468
MSFT	230	231	232	236	228

We can perform functions across the columns and rows, such as `colSum`:

In [19]:

```
colSums(stock.matrix)
```

Out[19]:

```
Mon
680
Tue
682
Wed
684
Thu
681
Fri
696
```

In [20]:

```
# Doesn't really make sense for stocks, but just to show how it works
rowSums(stock.matrix)
```

Out[20]:

```
GOOG
2266
MSFT
1157
```

We can also do other mathematical operations, check this [reference](#) for all functions available.

In [21]:

```
rowMeans(stock.matrix)
```

Out[21]:

GOOG

453.2

MSFT

231.4

Binding columns and rows

Let's go ahead and see how we can add columns and rows to a matrix, we can use the **cbind()** to bind a new column, and **rbind()** to bind a new row. For example, let's bind a new row with Facebook stock:

In [22]:

```
FB <- c(111,112,113,120,145)
```

In [23]:

```
tech.stocks <- rbind(stock.matrix,FB)
```

In [24]:

```
tech.stocks
```

Out[24]:

	Mon	Tue	Wed	Thu	Fri
GOOG	450	451	452	445	468
MSFT	230	231	232	236	228
FB	111	112	113	120	145

Now let's add an average column to the matrix:

In [25]:

```
avg <- rowMeans(tech.stocks)
```

In [26]:

```
avg
```

Out[26]:

GOOG

453.2

MSFT

231.4

FB

120.2

In [27]:

```
tech.stocks <- cbind(tech.stocks, avg)
```

In [28]:

```
tech.stocks
```

Out[28]:

	Mon	Tue	Wed	Thu	Fri	avg
GOOG	450.0	451.0	452.0	445.0	468.0	453.2
MSFT	230.0	231.0	232.0	236.0	228.0	231.4
FB	111.0	112.0	113.0	120.0	145.0	120.2

Great! We've gone over some basic matrix operations! Up next we will talk about selecting and indexing matrix elements!